

2080

Bachelor of Science in Computer Science and Information Technology

Course Title: **Simulation and Modeling**

Code No: **CSC328** Semester: **5th Semester**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. Why model of a system is built? What is static model? Differentiate between static and dynamic mathematical models in simulation.
2. What is storage in GPSS? Describe the blocks associated to storage in GPSS. A machine tool in a manufacturing shop is turning out parts at the rate of one every 5 minutes. As they are finished the parts go to an inspector who takes 10 ± 3 minutes to examine each one and rejects 10% of the parts. Represent the system in GPSS using the concept of facility and run the simulation for 500 parts.
3. What are the two main properties of random numbers? Test whether the 3rd, 7th, 11th, and so on numbers in the sequence in the following random number sample are auto-correlated. ($Z_{\alpha} = 0.05$ and $Z_{\alpha, \alpha, \alpha, \alpha} = 1.96$)
0.12 0.01 0.23 0.28 0.89 0.31 0.64 0.28 0.83 0.93 0.99 0.15 0.33 0.35 0.91 0.41 0.60 0.27 0.75 0.88
0.68 0.49 0.05 0.43 0.95 0.58 0.19 0.36 0.69 0.87

Section B

Attempt any Eight questions[8×5=40]

4. Explain Markov chain with suitable example.
5. What are steps involved in simulation study? Explain.
6. Generate 10 random integers using Linear congruential method where $m = 1000$, $a = 19$, $c = 6$, and $X_0 = 13$.
7. Explain different estimation methods which are used in simulation output analysis.
8. Define verification and validation. Explain the process of model verification in brief.
9. What is feedback system? Explain with example.
10. What is Calling Population? Explain Arrival and Service process in a queue.
11. Explain the Monte Carlo simulation method with example.
12. Write short notes: a) Non-stationary Poisson Process b) Poker Test